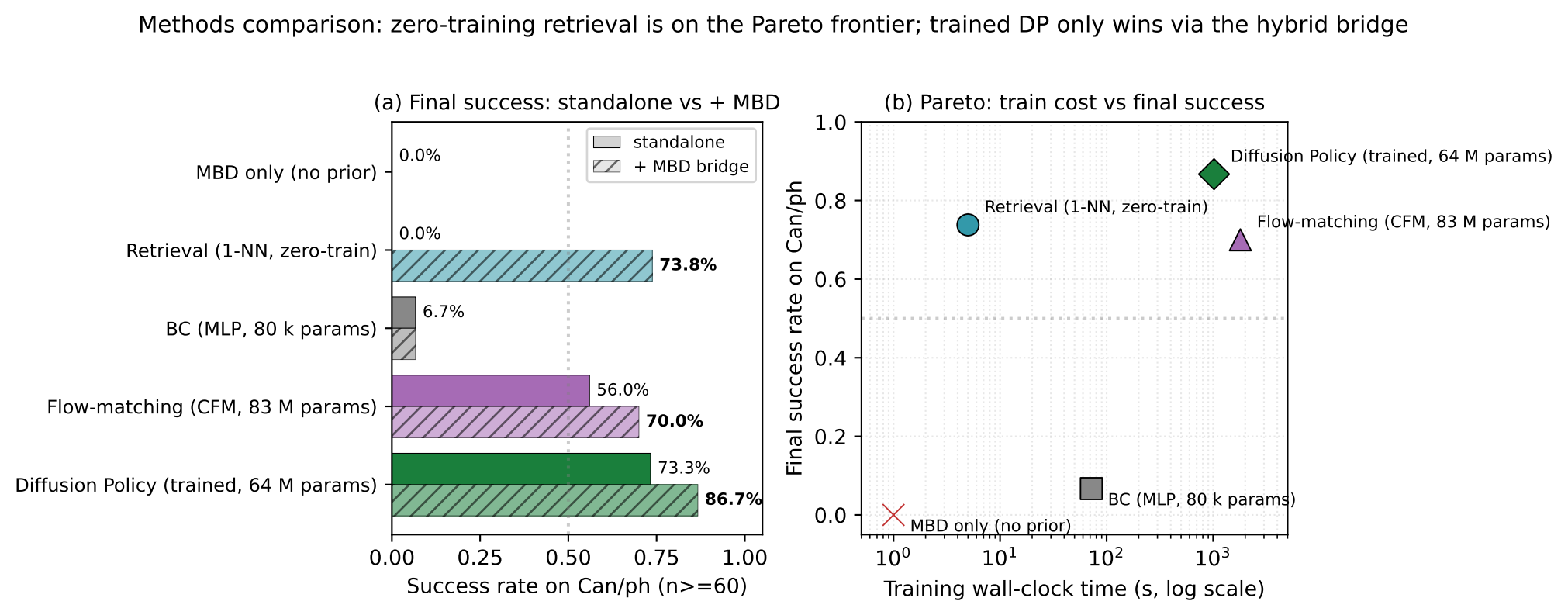
Methods comparison: zero-training retrieval is on the Pareto frontier; trained DP only wins via the hybrid bridge
(a) Final success: standalone vs + MBD
standalone
+ MBD bridge
MBD only (no prior)
Retrieval (1-NN, zero-train)
BC (MLP, 80 k params)
Flow-matching (CFM, 83 M params)
Diffusion Policy (trained, 64 M params)
0.0%
0.0%
73.8%
6.7%
56.0%
70.0%
73.3%
86.7%
0.00
0.25
0.50
0.75
1.00
Success rate on Can/ph (n>=60)
(b) Pareto: train cost vs final success
Final success rate on Can/ph
Training wall-clock time (s, log scale)
Diffusion Policy (trained, 64 M params)
Retrieval (1-NN, zero-train)
Flow-matching (CFM, 83 M params)
BC (MLP, 80 k params)
MBD only (no prior)